

SOFTWARE ENGINEERING PROJECT

KUru: RAG-Based Curriculum Understanding System

PLO–Skill–Career Navigator using LLM and Knowledge Graph

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Academic Year 2568

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Abstract

University curriculum documents — particularly the Thai Qualifications Framework for Higher Education document (มคอ.2) — are dense, formal texts written primarily for accreditation bodies rather than for students. Despite containing critical information about program learning outcomes (PLOs), required skills, and career pathways, these documents are rarely read or understood by the students they are intended to benefit.

KURu is a RAG-based web application that transforms the KU มคอ.2 document into an interactive, personalized career advisor. The system uses Retrieval-Augmented Generation (RAG) with Google Gemini API for semantic search and natural-language generation, and a Neo4j knowledge graph for structured PLO–Skill–Career relationship traversal. Students can query the curriculum in natural language with every answer grounded in the official document with page-level citations, explore their PLOs in plain language, visualize their skill gaps on a radar chart, receive career match recommendations with explainable gap analysis, and generate a personalized learning path toward their target career.

The system targets KU students across all year levels and is evaluated against a 50-question advisor-validated test set, targeting chatbot answer accuracy of 80% or above compared to a BM25 keyword-search baseline.

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Chapter 1

Introduction

1.1 Background

Higher education institutions design their degree programs around Program Learning Outcomes (PLOs) — formally defined statements that describe the knowledge, skills, and competencies that graduates of a program are expected to demonstrate. In Thailand, these outcomes are mandated and documented through the Thai Qualifications Framework for Higher Education (TQF), with the มคอ.2 document serving as the official curriculum specification for each program.

Despite the critical role PLOs play in shaping the educational journey of every student, there is a persistent and well-documented gap between the curriculum as designed and the curriculum as understood by students. The มคอ.2 document is written in formal academic Thai, structured to satisfy accreditation bodies, and rarely read by the students it is intended to benefit. As a result, students frequently progress through their degree without a clear understanding of what competencies they are building, what careers those competencies prepare them for, or what skills they still lack.

Recent advances in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) have created a new opportunity to bridge this gap. RAG architectures enable AI systems to answer questions grounded in specific documents with source citations, overcoming the hallucination problem of general-purpose LLMs. Combined with graph-based knowledge representations, these technologies make it possible to build systems that not only explain curriculum content in plain language, but also map that content to skills and career outcomes in a transparent and explainable way.

KURu (KU Curriculum Understanding) is a web-based application built on this foundation. It processes the official KU มคอ.2 document and transforms it into an interactive platform where students can query the curriculum, explore their PLOs, visualize skill gaps, and receive personalized career and learning path recommendations — all grounded in the official curriculum document.

1.2 Problem Statement

Within the Kasetsart University Computer Engineering program, three interconnected problems consistently affect student outcomes:

- **The formal Thai used in มคอ.2 is written for accreditation reviewers, not students. First-year students entering the program cannot interpret what PLO3 or PLO7 means in terms of skills they should develop or evidence they should produce.**PLO language is inaccessible.
- **Students select electives and plan their academic trajectory without a data-driven understanding of how each course contributes to their skill profile or moves them toward a specific career.**The connection between courses, skills, and careers is invisible.
- **Without a personal skill-gap analysis, students entering the job market or internship applications are unable to articulate which PLOs they have achieved or identify what specific skills they still need to develop.**Career planning is reactive rather than intentional.

These problems are not unique to KU — they reflect a structural limitation of outcome-based education documentation across Thai higher education — but KU provides a concrete and well-scoped context in which to address them.

1.3 Solution Overview

KURu transforms the static มคอ.2 curriculum document into an intelligent, interactive platform through two complementary AI architectures working in tandem.

The first is a Retrieval-Augmented Generation (RAG) pipeline. The มคอ.2 document is ingested, chunked by section, and indexed as semantic vector embeddings in a pgvector database using the Gemini text-embedding-004 model. When a student asks a question, the system retrieves the most semantically relevant curriculum passages and passes them to Gemini 2.5 Flash to generate a cited, plain-language answer. Every response is grounded in the official document — the system cannot answer from general knowledge, and any unanswerable question is explicitly flagged.

The second is a PLO–Skill–Career knowledge graph stored in Neo4j. All eight PLOs are extracted from the document and mapped to a 16-skill taxonomy and a career

dictionary with advisor approval. This graph enables deterministic, explainable career matching and learning path generation through Cypher graph traversal.

The system is deployed as a Next.js web application with a FastAPI backend, accessible to all KU students at a public URL at no cost.

1.3.1 Features

1. **A RAG-powered chatbot that answers student questions about the KU curriculum in Thai or English, with every response citing the relevant **หน้า.2** page and section.**Curriculum Q&A Chatbot:
2. **Each PLO is presented in three layers — a plain-language summary, a real-world example, and the original Thai text with page reference — alongside the skills it requires and the courses that address it.**PLO Explainer:
3. **A radar chart that visualises the student's self-assessed skill levels across 16 skills in four categories against the program's target levels, with a prioritised gap analysis.**Skill Map and Gap Analysis:
4. **Graph-based career recommendations showing match percentage, skill alignment, and specific gaps for each recommended career role.**Career Match Engine:
5. **A personalised, ordered sequence of courses, electives, and activities that close the student's skill gaps toward a chosen career target, with each step tagged to the PLOs it addresses.**Learning Path Generator:

1.4 Target User

KURu is designed for students enrolled in the Software and Knowledge Engineering (SKE) track within the Department of Computer Engineering at Kasetsart University. Four user personas capture the range of use cases:

- **who have just entered the program and need to understand what their PLOs mean in practical terms and what skills they are expected to develop across their degree.**First-year students (Year 1)

- **who are selecting electives and planning internship applications, and need to align their course choices with a specific career target.**Mid-program students (Year 2–3)
- **who are preparing for job applications and need to translate their academic experience into career-ready evidence, and identify any remaining skill gaps before graduation.**Final-year students (Year 4)
- **who can use the system's aggregated skill data (post-MVP) to identify which PLOs students are systematically underachieving across the cohort.**Faculty advisors

1.5 Terminology

Term	Definition
มคอ.2 (TQF2)	Thai Qualifications Framework Level 2 document — the official curriculum specification required by Thailand's Office of the Higher Education Commission.
PLO	Program Learning Outcome — a formally defined statement describing the knowledge, skills, or competency that a graduate of a program is expected to demonstrate.
RAG	Retrieval-Augmented Generation — an AI architecture combining semantic document retrieval with LLM generation, ensuring answers are grounded in a specific knowledge source.
Knowledge Graph (KG)	A structured data model that represents entities (PLOs, skills, careers) as nodes and their relationships as edges, enabling semantic queries and graph traversal.
pgvector	A PostgreSQL extension that stores and queries high-dimensional vector embeddings, enabling semantic similarity search within the Supabase relational database.
Skill Gap	The difference between a student's current self-assessed skill level and the target skill level required by a specific PLO or career role.
BM25	Best Match 25 — a classical keyword-based relevance ranking function used as the baseline retrieval method for comparison against the RAG semantic search approach.
Cypher	The declarative query language used to interact with Neo4j graph databases, used in KURu to traverse PLO–Skill–Career relationships.
Neo4j	An open-source graph database management system used to store and query the PLO–Skill–Career knowledge graph in KURu, self-hosted on Railway.

Chapter 2

Literature Review and Related Work

2.1 Competitor Analysis

Several existing systems address parts of the problem that KURu targets. The following analysis compares four relevant platforms across the dimensions most critical to KURu's value proposition: document grounding, PLO-specific support, skill gap analysis, career matching, and Thai language capability.

Feature	KURu (Proposed)	Knowva (KMITL)	ChulaGENIE (Chula)	General ChatGPT	Static มคอ.2 PDF
มคอ.2 grounding	RAG + citations	Partial — admission only	No — general LLM	No — hallucination risk	Raw text only
PLO plain-language explanation	Yes — structured layers	No	No	Partial — unpredictable	No
Skill gap analysis	Yes — radar chart	No	No	No	No
Career match engine	Yes — graph-based, explainable	Partial — general advice	No	No	No
Learning path generator	Yes — per student	No	No	No	No
Page-level citations	Yes — every answer	Unknown	No	No	N/A — static
Thai language support	Yes — Gemini native	Yes — Claude on AWS	Yes — Gemini multilingual	Partial	Yes — static text
Specific to KU SKE curriculum	Yes	No — KMITL only	No — Chula only	No	Yes — raw document

2.1.1 Knowva (KMITL)

Knowva is a generative AI virtual assistant developed by King Mongkut's Institute of Technology Ladkrabang (KMITL) in partnership with Amazon Web Services and DailiTech Co., Ltd., powered by Amazon Bedrock using Anthropic Claude. The system was designed specifically to answer prospective student queries about program information, admission processes, degree requirements, and career prospects. Following

its launch, Knowva handled over 6,000 student inquiries during an educational exhibition and achieved a 60 percent increase in student satisfaction compared to the previous year.

Knowva is the closest Thai university AI system to KURu in purpose, but differs in scope and depth. Knowva targets prospective students seeking admission information, not enrolled students navigating PLO-to-career pathways. It does not provide skill gap analysis, radar charts, or personalized learning path generation. KURu is informed by Knowva's demonstrated feasibility of Thai-language RAG in a university context and its success in reducing faculty time on repetitive queries.

2.1.2 ChulaGENIE (Chulalongkorn University)

ChulaGENIE is Chulalongkorn University's generative AI platform, launched in November 2024 in partnership with Google Cloud and built on Vertex AI using Gemini 1.5 Flash and Pro. It is the first university-wide generative AI application in Thailand, serving over 50,000 staff, faculty, and students. The platform supports multilingual document processing, custom AI agent creation, research assistance, academic advising, and administrative support.

ChulaGENIE is a general-purpose university AI platform rather than a curriculum-specific system. Its planned academic advisor agents most closely overlap with KURu's functionality, but they are not grounded in specific curriculum documents and do not provide PLO-specific explanations, skill radar charts, or structured career gap analysis. ChulaGENIE's architecture — specifically its use of Gemini on Vertex AI for Thai-language document processing — directly informs KURu's choice of Gemini 2.5 Flash as the generation and embedding model.

2.1.3 General-Purpose LLM Chatbots (ChatGPT, Gemini)

General-purpose LLMs such as OpenAI's ChatGPT and Google Gemini can answer curriculum-related questions when prompted, but are fundamentally unsuited for KURu's use case. They are not grounded in any specific university's มคอ.2 document, cannot provide page-level citations, and are prone to hallucination when asked about specific PLO codes, course requirements, or career mappings. KURu's RAG architecture is specifically designed to address this limitation.

2.2 Literature Review

KURu's design draws on four bodies of prior research: RAG architectures for educational applications, knowledge graphs in higher education, LLM-assisted curriculum modelling, and the combination of knowledge graphs with LLMs for explainable recommendations.

2.2.1 Retrieval-Augmented Generation in Education

The foundational RAG architecture was introduced by Lewis et al. (2020), who demonstrated that combining a dense retrieval component with a generative language model significantly improves factual accuracy on knowledge-intensive tasks compared to generation from parametric memory alone. This grounding property is central to KURu's design requirement that every chatbot answer must cite a specific [M00.2](#) page.

Thüs et al. (2024) developed OwlMentor, an RAG-based learning environment integrated into a university course over two semesters. Their study demonstrated that RAG systems can substantially increase accuracy — reporting up to 94% improvement in correct responses over unaugmented generation — and found that citation-linked responses were critical for student trust and engagement. This directly informs KURu's decision to display inline citations as badges rather than footnotes, making source verification a visible part of every answer.

A comprehensive survey of RAG in educational applications (2025) synthesises 51 studies across educational goals and contexts, identifying hallucination prevention and citation accuracy as the two most critical quality dimensions for educational RAG systems. The survey also highlights the particular challenge of processing domain-specific language — such as formal Thai academic text in [M00.2](#) — as a key factor in retrieval quality.

2.2.2 Knowledge Graphs for Curriculum and Learning Path Modelling

Abu-Rasheed et al. (2025) proposed an LLM-assisted pipeline for knowledge graph completion in higher education curriculum modelling. Their approach uses LLMs to extract fine-grained topics from lecture materials and map them to a domain model and user model within a knowledge graph, enabling personalized learning path recommendations. They found that LLM extraction is most effective when topics are contextually similar to existing graph content — a finding that directly informs KURu's decision to have the faculty advisor review and approve the PLO-to-skill mappings rather than relying on automated extraction alone.

A conceptual framework by researchers at Springer Nature (2025) introduces a knowledge graph-driven recommender for personalized course and curriculum path recommendation, arguing that knowledge graphs are particularly effective for organizing educational data because their structured, semantically meaningful representation enables coherent learning trajectory guidance. This validates KURu's use of Neo4j for the PLO–Skill–Career graph rather than a purely LLM-based recommendation approach.

2.2.3 Knowledge Graphs as Context for LLM Explanations

Abu-Rasheed et al. (2024) specifically investigated the use of knowledge graphs as context sources for LLM-generated explanations of learning recommendations, proposing a "conversational explainability" approach where knowledge graph data is injected into LLM prompts to reduce hallucination risk while maintaining natural dialogue. This is the precise architectural pattern KURu uses for its Career Match Engine: Neo4j provides the structured skill-gap data, and Gemini generates the natural-language explanation from that data rather than from general knowledge.

2.2.4 Thai Language Processing in Educational AI

The deployment experience of both Knowva and ChulaGENIE demonstrates that current-generation LLMs — specifically Anthropic Claude on AWS Bedrock (Knowva) and Google Gemini on Vertex AI (ChulaGENIE) — are sufficiently capable for Thai-language university document processing. ChulaGENIE's use of Gemini's multilingual capabilities for processing PDFs including Thai-language content with tables and charts directly establishes Gemini as a viable and well-validated choice for KURu's Thai มหาวิทยาลัย document processing pipeline.

2.2.5 Summary of Design Implications

The reviewed literature collectively supports the following key design decisions in KURu:

- RAG with mandatory citation is the appropriate architecture for curriculum Q&A, as demonstrated by OwlMentor and validated by the RAG-in-education survey.
- Knowledge graphs are the appropriate structure for PLO–Skill–Career mappings, providing explainable, deterministic outputs that are pedagogically preferable to probabilistic LLM recommendations.

- Human-in-the-loop validation of extracted graph content — here, faculty advisor review of PLO-to-skill mappings — is critical for accuracy in educational knowledge graphs, as shown by Abu-Rasheed et al. (2025).
- Gemini is a validated choice for Thai university document processing, established by ChulaGENIE's production deployment at scale.
- The combination of RAG (for unstructured narrative content) and knowledge graph traversal (for structured relational queries) is the most complete architecture for a system that must answer both open-ended curriculum questions and precise career-matching queries.

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Chapter 3

Requirement Analysis

3.1 Stakeholder Analysis

Stakeholders are individuals, groups, or entities that have an interest, concern, or stake in the KUru system — they can affect or be affected by the system's outputs and operation. Four primary stakeholders have been identified:

Stakeholder	Role / Interest	Concern / Impact
KU CPE / SKE Students (Year 1–4)	Primary users of all five features. They interact with the chatbot, explore PLOs, assess their skills, and generate learning paths.	Accuracy of AI answers, usefulness of career recommendations, relevance of learning paths, and data privacy of self-assessment scores.
Faculty Advisor	Provides the official มคอ.2 document, reviews and approves PLO-to-skill mappings in the knowledge graph, and validates the 50-question evaluation test set.	Accuracy of extracted PLO data, correctness of skill taxonomy, and integrity of the knowledge graph before student-facing deployment.
Department of Computer Engineering, KU	Institutional owner of the curriculum. Benefits from a system that improves student curriculum engagement and supports OBE accreditation evidence.	PDPA compliance, data security, reputational risk from incorrect AI-generated curriculum information.
Development Team (Project Authors)	Responsible for designing, building, testing, and deploying the system within the 4-month project timeline.	Technical feasibility, API cost management within free tiers, meeting the 80% accuracy target on time.

3.2 User Stories and Use Cases

User stories capture what each stakeholder needs from the system. The most critical user stories are expanded into use cases in Section 3.3.

3.2.1 Student User Stories

- As a first-year student, I want to ask what PLO3 means in plain language, so that I understand what skills I am expected to develop during my degree.
- As a third-year student, I want to see which skills I currently lack compared to the Software Developer career requirements, so that I can choose electives that close those gaps.

- As a final-year student, I want to generate a personalised learning path toward my target career, so that I can identify and address any remaining skill gaps before graduation.
- As a student, I want every chatbot answer to cite the specific ๙๙๐.2 page it came from, so that I can verify the information myself.
- As a student, I want to update my self-assessment skill scores and see my radar chart update immediately, so that I can track my progress over time.

3.2.2 Faculty Advisor User Stories

- As a faculty advisor, I want to review and approve the PLO-to-skill mappings before they are made available to students, so that I can ensure the knowledge graph reflects the intended curriculum design.
- As a faculty advisor, I want to validate the 50-question chatbot test set so that the accuracy evaluation reflects real curriculum questions that students would ask.

3.2.3 Key Use Cases

UC-01: Ask a Curriculum Question (Chatbot)

Actor: Student

Precondition: ๙๙๐.2 document has been ingested and indexed in pgvector.

Main Flow:

6. Student enters a question in Thai or English in the chat interface.
7. System embeds the query using Gemini text-embedding-004.
8. System retrieves the top-5 most semantically relevant curriculum chunks from pgvector.
9. System passes the chunks and question to Gemini 2.5 Flash with a strict citation instruction.
10. System displays the generated answer with an inline ๙๙๐.2 page citation badge.

Alternative Flow: If no relevant chunk is found, the system responds: "This information was not found in the curriculum document."

Postcondition: Student has received a cited answer grounded in the official curriculum document.

UC-02: Explore a PLO in Plain Language

Actor: Student

Precondition: PLO data has been extracted and stored in Supabase and Neo4j.

Main Flow:

11. Student navigates to the PLO Explainer page.
12. Student selects a PLO (e.g. PLO3) from the list.
13. System retrieves the PLO record from Supabase and the associated skills from Neo4j.
14. System displays: (a) a one-sentence plain-language summary, (b) a concrete real-world example, (c) the original Thai text with page reference, and (d) the linked skill nodes.

Postcondition: Student understands what the PLO means in practical terms and what skills it requires.

UC-03: View Skill Map and Gap Analysis

Actor: Student

Precondition: Student has a profile in the system; skill taxonomy and PLO-to-skill mappings are stored in Neo4j.

Main Flow:

15. Student navigates to the Skill Map page.
16. Student adjusts sliders to self-assess current skill levels across 16 skills.

17. System calculates gaps between self-assessed levels and program target levels from Neo4j.
18. System renders a radar chart showing current vs. target skill levels.
19. System highlights the 2–3 most critical skill gaps with suggested courses to address each.

Postcondition: Student has a visual understanding of where their skill profile stands relative to program targets.

UC-04: Get Career Match Recommendations

Actor: Student

Precondition: Student skill levels have been set (via self-assessment); Career dictionary and skill requirements are stored in Neo4j.

Main Flow:

20. Student navigates to the Career Match page.
21. System queries Neo4j to compute a match percentage for each career role based on the student's current skill levels.
22. System ranks and displays the top 3–5 career matches with: (a) match percentage, (b) skills that already meet career requirements, and (c) specific skills still needed.
23. Student can click any career card to see a Gemini-generated plain-language explanation of the recommendation.

Postcondition: Student understands which careers they are most aligned with and exactly what skills they need to develop to reach their target career.

UC-05: Generate a Personalised Learning Path

Actor: Student

Precondition: Career Match has been viewed; student has selected a target career.

Main Flow:

- 24. Student selects a target career from the Career Match page.
- 25. System queries Neo4j for the skill gaps between the student's current levels and the career's required levels.
- 26. System generates an ordered list of recommended courses, electives, and activities that close each gap.
- 27. Each step is tagged with the PLO it addresses, so the student can see how each action connects back to the curriculum's intended outcomes.
- 28. Student can also click "Generate with AI" to receive a Gemini-generated personalised narrative path.

Postcondition: Student has an actionable, ordered roadmap from their current skill profile to their target career.

3.3 Use Case Diagram

The use case diagram below illustrates all actors and their relationships to the five primary use cases of the KUru system. The Student actor interacts with all five features directly. The Faculty Advisor actor interacts with the system during the data preparation and validation phase (pre-deployment) rather than through the student-facing UI.

«Actor» Student	UC-01 Ask a Curriculum Question (Chatbot)	KUru System
	UC-02 Explore PLO in Plain Language	
	UC-03 View Skill Map and Gap Analysis	
	UC-04 Get Career Match Recommendations	«Actor» Faculty Advisor
	UC-05 Generate Personalised Learning Path	UC-06 Review & Approve PLO–Skill Mappings

Note: The above table is a schematic representation of the use case diagram. A formal UML diagram will be produced in the project documentation using draw.io and included in the final submission.

3.4 User Interface Design

KUru's interface follows a single-page application pattern with a persistent left navigation sidebar and a main content area. The design prioritises clarity and information density appropriate for academic users. The interface was prototyped as an interactive AI-powered mock (built with Next.js components) and validated with five KU CPE students during the requirements phase.

3.4.1 Navigation Structure

The application is divided into five primary screens, accessible via the sidebar:

29. Curriculum Chatbot — conversational interface with quick-chip suggested questions, chat history, and inline citation badges on every AI response.
30. PLO Explainer — grid of PLO cards, each expandable to reveal plain-language summary, real-world example, original Thai text, and linked skill pills.
31. Skill Map — split view: radar chart on the left, skill sliders and gap badges on the right. Updates in real time as sliders are adjusted.
32. Career Match — grid of career cards showing match percentage bar, explanation, and skill gap list. Each card has an AI explain button powered by Gemini.
33. Learning Path — dropdown to select a target career, followed by a vertical timeline of recommended courses tagged with PLOs. "Generate with AI" button produces a Gemini-narrated personalised path.

3.4.2 Key UI Design Decisions

- **Every chatbot answer displays the ๓๓๓.2 page citation as a highlighted badge (amber background) inline with the answer text — not in a footnote — making source verification immediately visible.** Citation as first-class UI element:
- **The skill radar chart re-renders on every slider change rather than on form submission, giving students immediate visual feedback and encouraging exploration.** Radar chart with real-time updates:

- **Career match percentages are always accompanied by a breakdown of which skills already meet requirements and which do not. A percentage alone is never displayed without its explanation.** Explainability over opacity:
- **All chatbot interactions accept Thai and English input. The system detects the language of the query and responds in the same language.** Bilingual support:
- **PLO cards show only the title and skill pills by default. The full three-layer explanation is revealed on click, preventing information overload for first-year students while providing depth for advanced users.** Progressive disclosure in PLO Explainer: